Diagnostic check: six questions, 10 minutes

Answer without consulting the course. Indicate the questions for which assistance is required. diagnostic check is not graded.

1. For $A(1; 2)$ and $B(4; -2)$, calculate $\overrightarrow{AB}$.

2. Calculate the norm of $(3; -4)$.

3. Simplify $\overrightarrow{AB} + \overrightarrow{BC}$.

4. Develop $(a + b)^2$.

5. In a right triangle, an angle has adjacent side 3 and hypotenuse 5. Find its cosine.

6. The midpoint of $A(-2; 1)$ and $B(4; 3)$?

After the diagnostic check. Retrieve the concepts indicated with the teacher; repeat a question and then explain the rule used.

Speaking and reasoning. Compare, on the same problem, coordinates and vector calculation without reference. Explain why your method uses the data most directly.

Search before formalizing

Build three M points such as $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ for an AB segment of 6 cm. Conjecture the locus then produce evidence that includes A and B.

Organization. Individual research 5 min, confrontation 2 5 min, pooling 5 min.
Keep a trace of a test, even if not completed. A conjecture must then be justified.